

WASP-36b

R-1278

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_190213 · created 2026-07-31T14:38:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458527.7238 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.047
Transit depth (Rp/Rs)^2	2.24 +/- 1.41 [%]
Orbital Inclination (inc)	87.45 +/- 2.83
Airmass coefficient 1 (a1)	1.068 +/- 0.062
Airmass coefficient 2 (a2)	-0.048 +/- 0.051
Scatter in the residuals of the lightcurve fit is	5.77 %
Best Comparison Star	#1 - [214, 149]
Optimal Aperture	2.29
Optimal Annulus	8.45
Transit Duration (day)	0.0912 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.047 vs 0.1368 (deviation 0.27σ)
Duration fit vs published	0.0912 d vs 0.0758 d (deviation 1.76σ)
Mid-time O–C	-7.9 min
Depth SNR	3.1
Residual scatter	5.77 %

Light curve

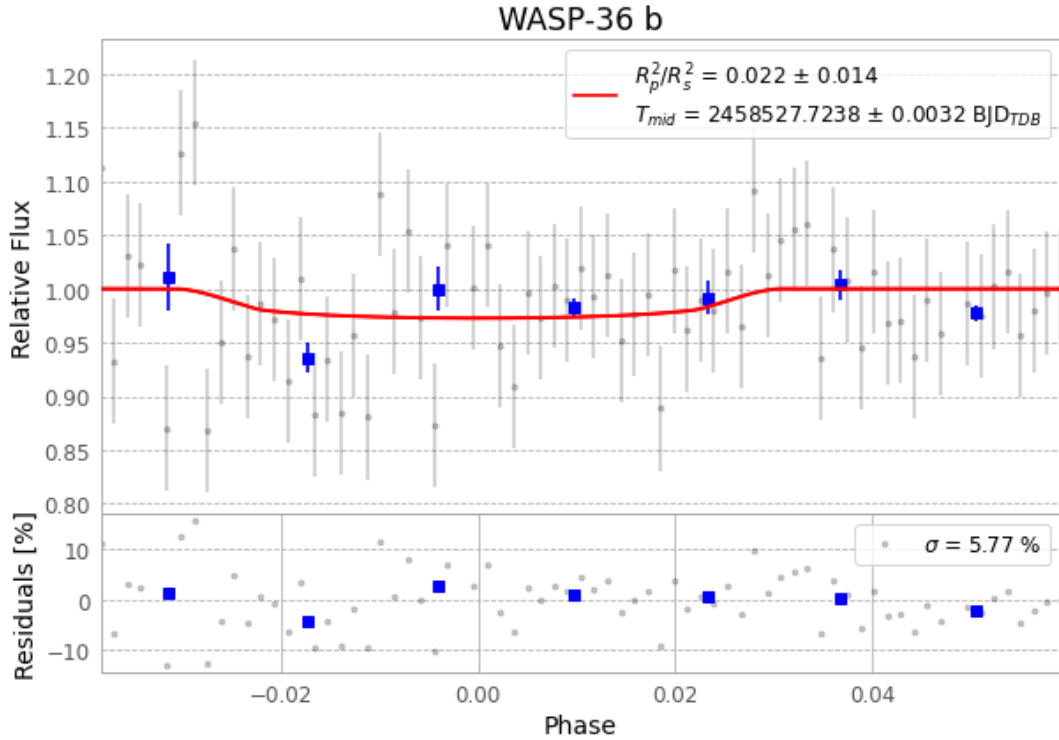


Figure 1 — FinalLightCurve_WASP-36 b_2019-02-12.png (SHA-256 4578e501ab018f7d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 130], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7c7b5a0ae019a0df...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-36190213034812.FITS → WASP-36190213072412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190213.log (5da525324f4e...), FinalLightCurve_WASP-36 b_2019-02-12.png (4578e501ab01...), FinalLightCurve_WASP-36 b_2019-02-12.csv (96c3262180b9...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:38Z.